21. ROS2 URDF model

1. Introduction to URDF

The modeling method in ROS is called URDF, which stands for Unified Robot Description Format. It can not only clearly describe the robot model itself, but also describe the external environment of the robot. URDF model files use the XML format.

SolidWorks model to URDF plugin: [sw_urdf_exporter - ROS Wiki]

2. Composition of the robot

During the process of modeling and describing a robot, we first need to be familiar with the composition and parameters of the robot. For example, a robot is generally composed of four major parts: **the hardware structure, the drive system, the sensor system and the control system**. Some of the common robots on the market, whether mobile robots or robotic arms, can all be broken down according to these four major parts.



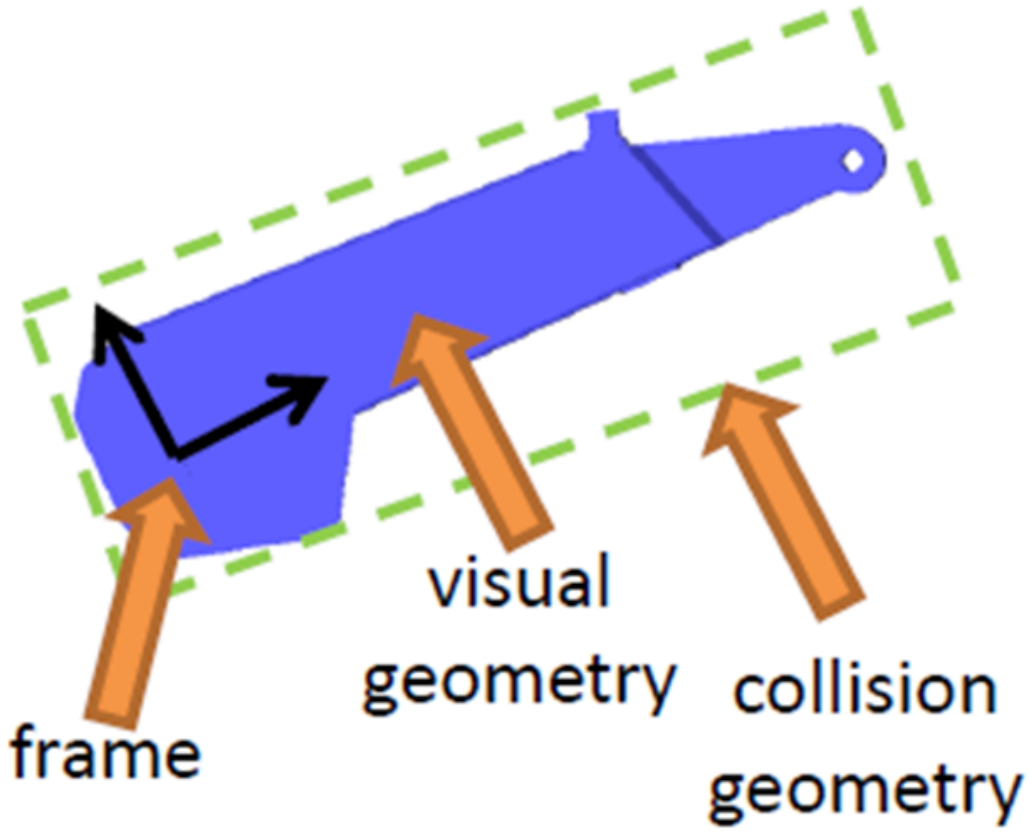
- The hardware structure refers to the devices that you can really see, such as the chassis, the housing and the motors;
- The drive system is the device that drives these devices to work properly, such as the motor driver, the power management system, etc.;
- The sensor system includes the encoder on the motor, the onboard IMU, the installed camera, the radar, etc., which makes it convenient for the robot to sense its own state and the external environment;
- The control system is the main carrier of our development process, generally a computing platform such as a Raspberry Pi or a computer, as well as the operating system and the application software inside it.

The process of robot modeling is actually a similar idea: describe every part of the robot clearly with a modeling language and then combine them together.

3. URDF syntax

3.1. Description of the link

The tag is used to describe the appearance and the physical properties of a rigid body of the robot. The appearance includes the size, the color and the shape, and the physical properties include the mass, the inertia matrix, the collision parameters, etc.



Taking this robotic arm link as an example, its link description is as follows:

```
<link name="link_4">
    <visual>
        <geometry>
            <mesh filename="link_4.stl"/>
        </geometry>
        <origin xyz="0 0 0" rpy="0 0 0" />
    </visual>
    <collision>
        <geometry>
            <cylinder length="0.5" radius="0.1"/>
        </geometry>
        <origin xyz="0 0 -0.05" rpy="0 0 0" />
    </collision>
</link>
```

The name in the link tag indicates the name of the link. We can customize it, and this name will be used when the joint connects the links in the future.

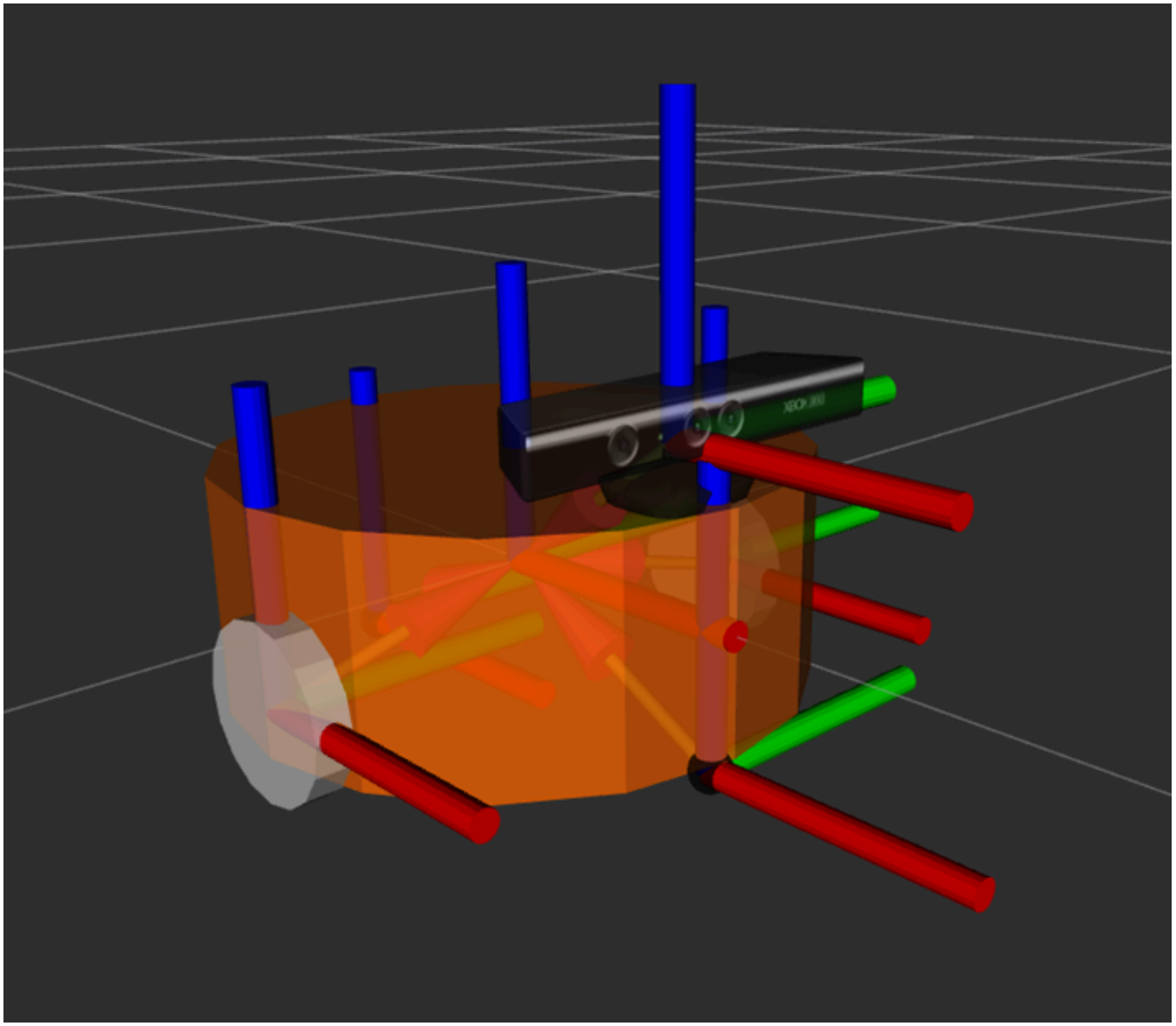
The part in the link is used to describe the appearance of the robot, for example:

- indicates the geometric shape. The inside it calls a blue appearance that was designed in advance in 3D software, that is, this stl file, which looks the same as the real robot.
- indicates the offset of the coordinate system relative to the initial position, which is the translation along the x, y and z directions, and the roll, pitch and raw rotations. If no offset is needed, all of them are 0.

The second part, , describes the collision parameters. The content inside it seems to be the same as , and it also has and . Although they look the same, the difference is still relatively large.

- The part focuses on describing what the robot looks like, that is, the visual effect;
- The part describes the state of the robot during its motion, for example, how the robot is considered to be in collision with the outside world.

In this robot model, the blue part is described by . During actual control, such a complex appearance requires high computing power when the collision detection is calculated. To simplify the calculation, we simplify the model used for the collision detection into the cylinder framed in green, which is the shape described by inside . The coordinate system offset is similar and can describe the offset of the center of mass of the rigid body.



If it is a mobile robot, the link can also be used to describe the body of the car, the wheels and other parts.

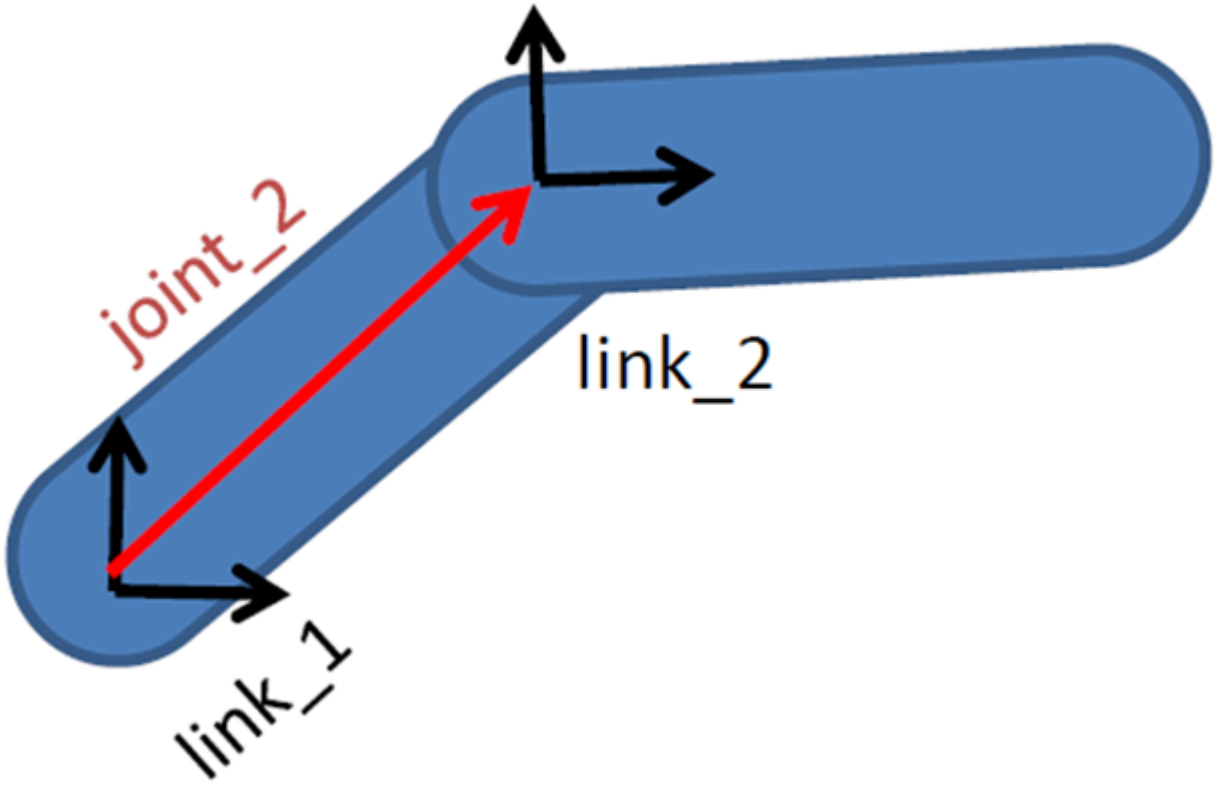
3.2. Description of the joint

The rigid bodies in the robot model can finally produce a relative motion only after they are connected by the joints.

There are six motion types for the joints in URDF.

关节类型	描述
continuous	旋转关节，可以围绕单轴无限旋转
revolute	旋转关节，类似于continuous，但是有旋转的角度极限
prismatic	滑动关节，沿某一轴线移动的关节，带有位置极限
fixed	固定关节，不允许运动的特殊关节
floating	浮动关节，允许进行平移、旋转运动
planar	平面关节，允许在平面正交方向上平移或者旋转

1. continuous, which describes rotational motion and can rotate infinitely around a certain axis. For example, the wheels of a car belong to this type.
2. revolute, which is also a revolute joint. The difference from the continuous type is that it cannot rotate infinitely but has an angle limit. For example, the two links of a robotic arm belong to this motion.
3. prismatic, which is a sliding joint. It can translate along a certain axis and also has a position limit. Generally, a linear motor uses this motion mode.
4. fixed, a fixed joint. It is the only joint that does not allow any motion, but it is still used quite frequently. For example, a camera link is installed on the robot and its relative position does not change. The connection type used in this case is Fixed.
5. Floating is the floating joint, and the sixth type, planar, is the planar joint. These two are used relatively less.

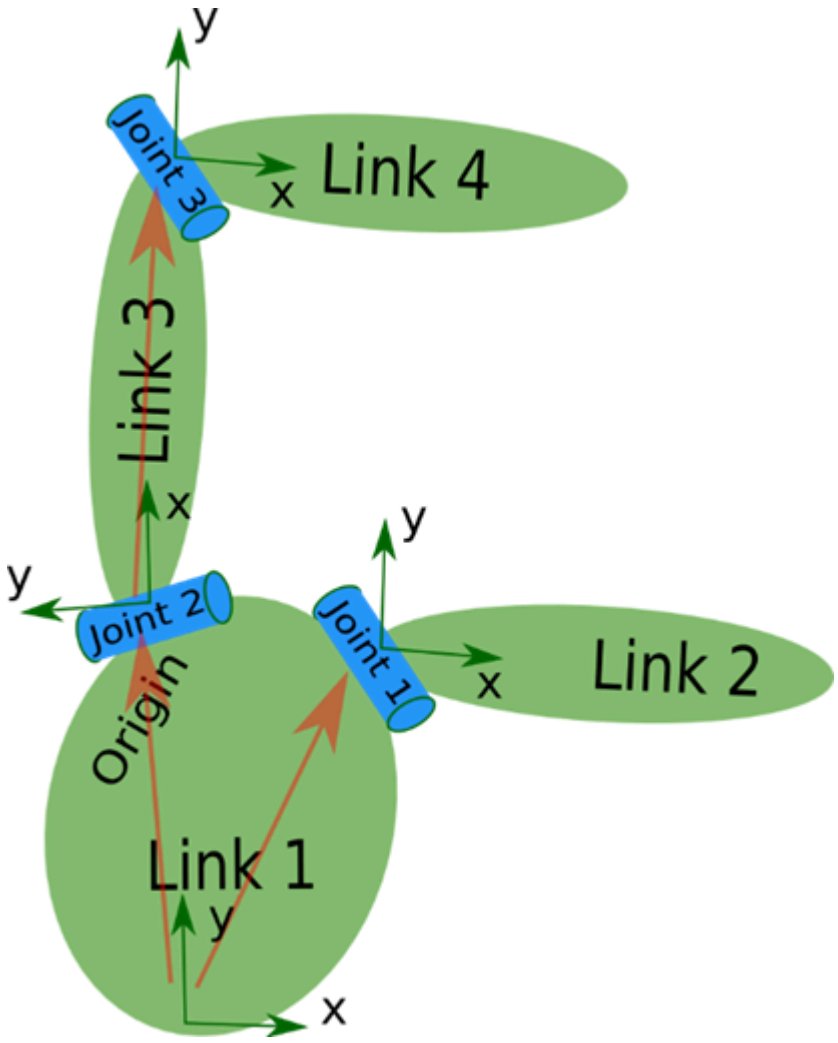


In a URDF model, every link is described with such a piece of xml content, for example what the name of the joint is and which motion type it is.

```
<joint name="joint_2" type="revolute">
    <parent link="link_1"/>
    <child link="link_2"/>
    <origin xyz="0.2 0.2 0" rpy="0 0 0"/>
    <axis xyz="0 0 1"/>
    <limit lower="-3.14" upper="3.14" velocity="1.0"/>
</joint>
```

- parent tag: describes the parent link;
- child tag: describes the child link, and the child link moves relative to the parent link;
- origin: indicates the relationship between the coordinate systems of the two links, which is the red vector in the figure. It can be understood as how the two links should be mounted together;
- axis indicates the unit vector of the motion axis of the joint. For example, if z is equal to 1, it means that this rotational motion is carried out around the positive direction of the z axis;
- limit indicates some restrictions on the motion, such as the minimum position, the maximum position and the maximum speed, etc.

4. Complete robot model



Finally, all the link and joint tags complete the description and the combination of every part of the robot. Putting them all into a single robot tag forms the complete robot model.

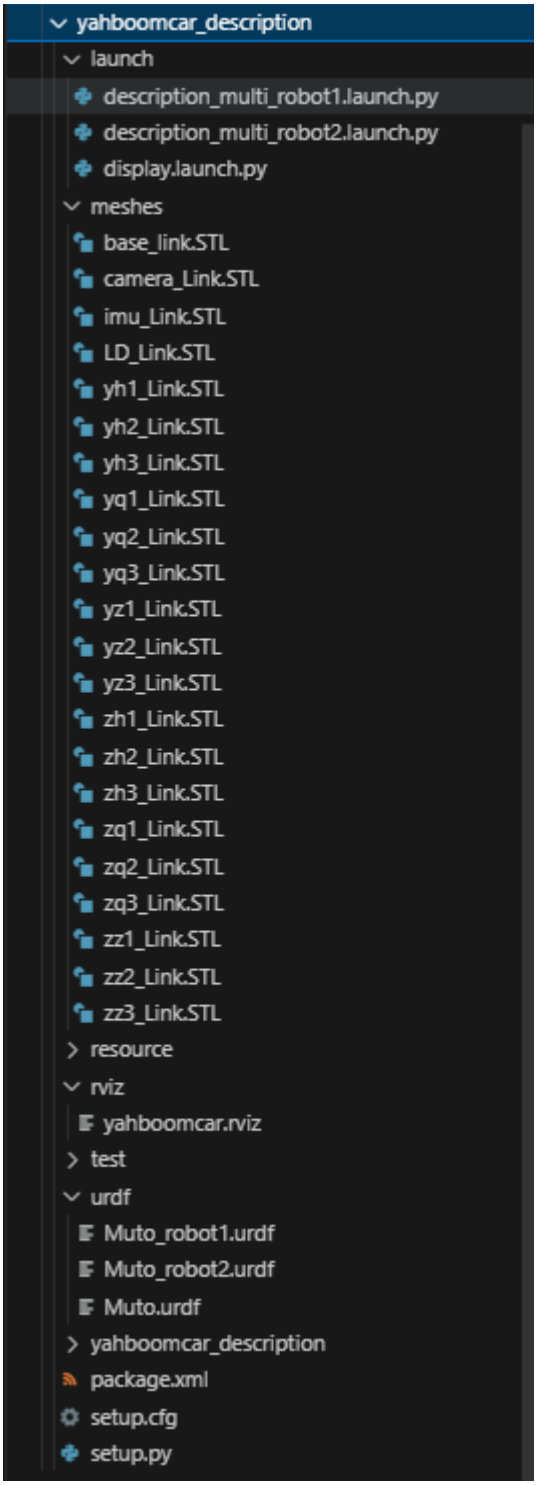
```
<robot name="<name of the robot>">
    <link> ..... </link>
    <link> ..... </link>

    <joint> ..... </joint>
    <joint> ..... </joint>
</robot>
```

So when you are looking at a certain URDF model, do not rush to look at the details of each block of code. First, look for the link and the joint to see which parts this robot is composed of. After you understand the overall picture, then look at the details.

4.1. Creating the robot model

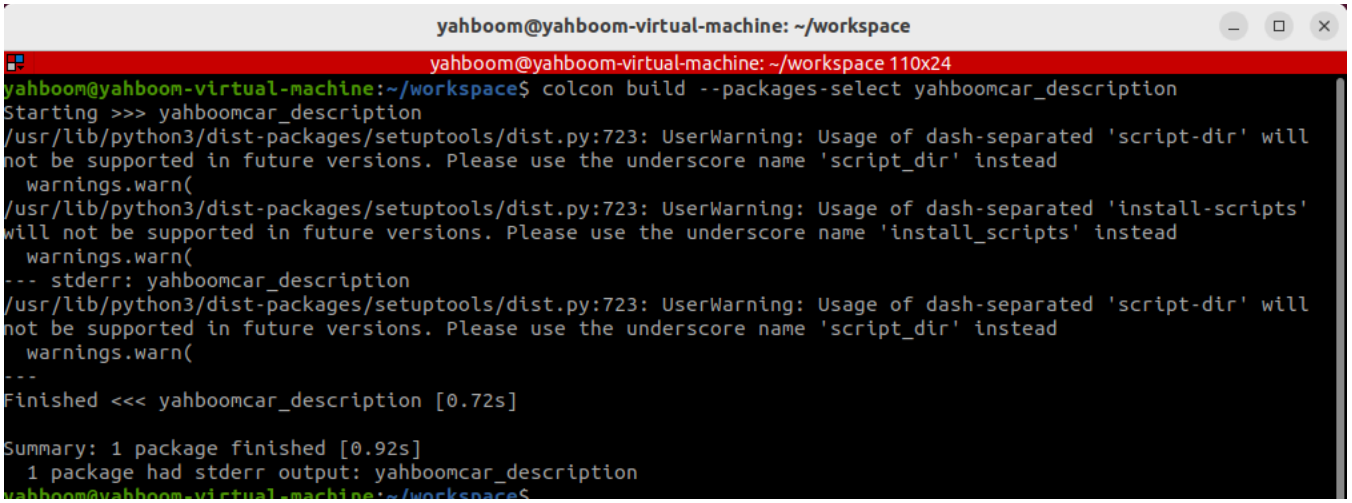
Taking the muto model as an example, copy the `yahboomcar_description` function package in the folder of this lesson into the src directory of the workspace:



- urdf: stores the URDF or xacro file of the robot model
- meshes: stores the model rendering files referenced in the URDF
- launch: saves the relevant startup files
- rviz: saves the rviz configuration file

Then compile the function package

```
colcon build --packages-select yahboomcar_description
```



4.2. Model visualization result

- Refresh the environment variables and then run the startup command

```
ros2 launch yahboomcar_description display.launch.py
```

Afterwards rviz is started automatically and displays the robot model:

